ESD Question Bank

Unit 1 — MCQ Question Bank with Answers

Question and options on the left, answer with a one-line reason on the right. Original question numbering and answer key kept.

Odd / Even

Question	Answer
1. Pick the odd one out 3, 5, 11, 14, 17, 21 a) 21 b) 17 c) 14 d) 3	**c) 14** 14 is the only even number; all the others are odd.
2. Pick the odd one out 331, 482, 551, 263, 383, 362, 284 a) 263 b) 383 c) 331 d) 551	**b) 383** In every other number, middle digit = first × last digit (3×1=3, 4×2=8…); for 383, 3×3 ≠ 8.
12. Pick the odd one out 8, 27, 64, 100, 125, 216, 343 a) 27 b) 100 c) 125 d) 343	**b) 100** All others are perfect cubes (2^3 to 7^3); 100 is not.
23. Pick the odd one out 10, 25, 45, 54, 60, 75, 80 a) 10 b) 45 c) 54 d) 75	**c) 54** All others are multiples of 5.
34. Pick the odd one out 396, 462, 572, 427, 671, 264 a) 396 b) 427 c) 671 d) 264	**b) 427** In the others, middle digit = sum of the outer digits (3+6=9, 4+2=6…); for 427, 4+7 ≠ 2.
36. Pick the odd one out6, 9, 15, 21, 24, 28, 30 a) 28 b) 21 c) 24 d) 30	**a) 28** All others are multiples of 3.
37. Pick the odd one out1, 4, 9, 16, 23, 25, 36 a) 9 b) 23 c) 25 d) 36	**b) 23** All others are perfect squares.
38. Pick the odd one out 1, 4, 9, 16, 20, 36, 49 a) 1 b) 9 c) 20 d) 49	**c) 20** All others are perfect squares.
39. Pick the odd one out 2, 5, 10, 17, 26, 37, 50, 64 a) 50 b) 26 c) 37 d) 64	**d) 64** Series is $n^2 + 1$ (2, 5, 10, 17…); after 50 comes 65, not 64.
40. Pick the odd one out 10, 14, 16, 18, 21, 24, 26 a) 26 b) 24 c) 21 d) 18	**c) 21** 21 is the only odd number.

H.C.F & L.C.M

No question in the source was identified as a direct match for this topic.

Square Roots & Cube Roots

No question in the source was identified as a direct match for this topic.

Average

Question	Answer
19. The average of 5 consecutive numbers A,B,C,D and E is 52. What is the sum of B and E? a) 102 b) 103 c) 104 d) 105	**d) 105** Numbers are 50, 51, 52, 53, 54 → B + E = 51 + 54 = 105.
24. Average age of students of an adult school is 40 years. 120 new students whose average age is 32 years joined the school. As a result the average age is decreased by 4 years. Find the number of students in the school after joining of the new students. a) 1200 b) 120 c) 360 d) 240	**d) 240** 40x + 120×32 = 36(x + 120) → x = 120; after joining: 120 + 120 = 240.
25. The average age of 5 men is 20 years. Their average age is increased by 1 when a new man joined them. Find the age of the new man a) 27 b) 28 c) 26 d) None of these	**c) 26** New total 6 × 21 = 126; old total 5 × 20 = 100 → new man = 26.
326. 50 men took a dip in a water tank 40 m long and 20 m broad on a religious day. If the average displacement of water by a man is 4 cu.m , then the rise in the water level in the tank will be: a) 20 cm b) 25 cm c) 30 cm d) 35 cm	**b) 25 cm** Displaced = 50 × 4 = 200 m^3; rise = 200 ÷ (40 × 20) = 0.25 m = 25 cm.

Percentage

Question	Answer
14. In an election contested by two parties, Party D secured 12% of the total votes more than Party R. If party R got 132,000 votes, by how many votes did it lose the election? a) 32000 b) 36000 c) 40000 d) 44000	**b) 36000** R = 44% of total = 132000 → total = 300000; margin = 12% = 36000.
16. A number on subtracting 15 from it, reduces to its 80%. What is 40% of that number? a) 25 b) 40 c) 30 d) cannot be determined	**c) 30** x − 15 = 0.8x → x = 75; 40% of 75 = 30.
18. The cost of a radio during 1980 was Rs. 120. In 1988, the price had increased to Rs 180. What was the percentage increase in the cost of the radio? a) 20% b) 30% c) 40% d) 50%	**d) 50%** (180 − 120) ÷ 120 × 100 = 50%.
272. The ratio of expenditure and savings is 3 : 2 . If the income increases by 15% and the savings increases by 6% , then by how much percent should his expenditure increases? a) 21 b) 22 c) 23 d) 24	**a) 21** Income 5 → 5.75, savings 2 → 2.12, so expenditure 3 → 3.63: increase 0.63/3 = 21%.
330. Find the number of bricks, each measuring 24 cm x 12 cm x 8 cm, required to construct a wall 24 m long, 8m high and 60 cm thick, if 10% of the wall is filled with mortar? a) 45000 b) 23000 c) 56000 d) 35000	**a) 45000** Brick volume = 90% × (2400×800×60) = 103,680,000 cm^3; ÷ (24×12×8 = 2304) = 45000.

Question	Answer
338. Dean has a cardboard box whose length, breadth and height are in the ratio 1:2:3. He makes a new box such that the length, breadth and height got increased by 100%, 200% and 200% respectively. How much less is volume of old box than the new box? a) 12 times less b) 15 times less c) 17 times less d) 20 times less	**c) 17 times less** Old 1×2×3 = 6; new 2×6×9 = 108; difference 102 = 17 × old volume.
358. If each edge of a cube is increased by 50%, find the percentage increase in Its surface area a) 125% b) 150% c) 175% d) 110%	**a) 125%** Area scales as edge²: $1.5^2 = 2.25 \rightarrow 125\%$ increase.

Ratios & Proportions

Question	Answer
8. The ratio of dogs to cats in a pet store is 3 is to 4. What is the number of dogs in the store if there are 20 cats? a) 10 b) 15 c) 20 d) 25	**b) 15** 4 parts = 20 → 1 part = 5; dogs = 3 × 5 = 15.
13. What sum of money is to be divided among 3 men in the ratio 3 : 4 : 5 so that the third man receives Rs 10 only ? a) 56 b) 84 c) 120 d) 24	**d) 24** 5 parts = 10 → 1 part = 2; total 12 parts = 24.
22. The ratio of ages of two persons is 4 : 7 and one is 30 years older than the other. Find the sum of their ages. a) 210 b) 110 c) 90 d) 140	**b) 110** Difference 3 parts = 30 → 1 part = 10; sum 11 parts = 110.
33. The ratio of the age of a gentleman and his wife is 4 : 3. After 4 years this ratio will be 9 : 7. If at the time of their marriage the ratio was 5 : 3, how many years ago they were married? a) 10 b) 8 c) 12 d) 15	**c) 12** $(4x+4)/(3x+4) = 9/7 \rightarrow x = 8$: ages 32, 24; $(32-t)/(24-t) = 5/3 \rightarrow t = 12$.
54. Rs. 8400 is divided among A, B, C and D in such a way that the shares of A and B, B and C, and C and D are in the ratios of 2:3, 4:5 and 6:7 respectively. The share of A is a) 1280 b) 8400 c) 8210 d) 1320	**a) 1280** A:B:C:D = 16:24:30:35 (sum 105); A = 8400 × 16/105 = 1280.
55. The ratio of the present age of father to that of son is 7:2. After 10 years their ages will be in the ratio of 9:4. The present ages of the father is a) 35 b) 45 c) 30 d) 50	**a) 35** $(7x+10)/(2x+10) = 9/4 \rightarrow x = 5$; father = 35.
56. In a library, the ratio of number of story books to that of non-story books was 4:3 and total number of story books was 1248. When some more story books were bought, the ratio became 5:3. Find the number of story books bought. a) 312 b) 321 c) 936 d) 1560	**a) 312** Non-story = 936; new story = 936 × 5/3 = 1560; bought = 1560 − 1248 = 312.
57. Ajay and Raj together have Rs. 1050. On taking Rs. 150 from Ajay, Ajay will have same amount as what Raj had earlier. Find the ratio of amounts with Ajay and Raj initially. a) 2:3 b) 3:2 c) 4:3 d) 1:3	**c) 4:3** A + R = 1050, A − 150 = R → A = 600, R = 450 → 4:3.

Question	Answer
58. Price of each article of type P, Q, and R is Rs. 300, Rs. 180 and Rs. 120 respectively. Suresh buys articles of each type in the ratio 3:2:3 in Rs. 6480. How many articles of type Q did he purchase? a) 8 b) 14 c) 20 d) 25	**a) 8** 900x + 360x + 360x = 1620x = 6480 → x = 4; Q = 2x = 8.
59. The ratio of market prices of wheat and paddy is 2:3 and the ratio of quantities consumed in a family is 5:4. Find the ratio of expenditure of wheat and paddy a) 5:6 b) 6:5 c) 1:3 d) 4:5	**a) 5:6** Expenditure = price × quantity = (2×5):(3×4) = 10:12 = 5:6.
60. The ratio of numbers of girls and boys participating in sports of a school is 4:5. If the number of girls is 212, determine the number of boys participating in the sports a) 265 b) 256 c) 251 d) 263	**a) 265** 212 ÷ 4 × 5 = 265.
61. The three numbers are in the ratio 1/2 : 2/3 : 3/4. The difference between greatest and smallest numbers is 36. Find the numbers a) 72, 84, 108 b) 60, 72, 96 c) 72, 84, 96 d) 72, 96, 108	**a) 72, 84, 108** Answer as per the given key.
62. If A:B = 2:3, B:C = 4:5 and C:D = 6:7, then A:B:C:D is a) 18:24:30:35 b) 16:24:30:35 c) 16:22:30:35 d) 16:24:15:35	**b) 16:24:30:35** Make B and C common: A:B:C = 8:12:15, then ×2 with C:D → 16:24:30:35.
64. Ratio of two numbers is 3:8. On adding 5 to both numbers, the ratio becomes 2:5. Which is the smaller number out of the two? a) 64 b) 120 c) 45 d) 105	**c) 45** (3x+5)/(8x+5) = 2/5 → x = 15; smaller = 45.
196. The ratio between the length and the breadth of a rectangular park is 3 : 2. If a man cycling along the boundary of the park at the speed of 12 km/hr completes one round in 8 minutes, then the area of the park (in sq. m) is: a) 15360 b) 153600 c) 30720 d) 307200	**b) 153600** Perimeter = 200 m/min × 8 = 1600 m = 10x → x = 160; area = 480 × 320 = 153600.

Partnership

Question	Answer
43. A and B started a business by investing Rs.4000/- and Rs.5000/- respectively. Find the A's share out of a total profit of Rs.1800 a) Rs.1000/- b) Rs.1800/- c) Rs.800/- d) Rs.400/-	**c) Rs.800/-** Ratio 4:5 → A = 1800 × 4/9 = 800.
44. A, B, C subscribe Rs. 50,000 for a business. A subscribes Rs. 4000 more than B and B Rs. 5000 more than C. Out of a total profit of Rs. 35,000, A receives a) Rs.8400/- b) Rs.11,500/- c) Rs.11,000/- d) Rs.14,700/-	**d) Rs.14,700/-** C = 12000, B = 17000, A = 21000; A = 35000 × 21/50 = 14700.
45. A,B and C started a business by investing Rs.2000/- , Rs.3000/- and Rs.4000/- respectively. Total profit is Rs.4500/-. Find the A's share? a) Rs.2000/- b) Rs.1500/- c) Rs.1000/- d) Rs.4500/-	**c) Rs.1000/-** Ratio 2:3:4 → A = 4500 × 2/9 = 1000.

Question	Answer
46. A,B and C started a business by investing Rs.1400/- , Rs.4200/- and Rs.7000/- respectively. Total profit is Rs.8100/-. Find the C's share? a) Rs.900/- b) Rs.1800/- c) Rs.2700/- d) Rs.4500/-	**d) Rs.4500/-** Ratio 1:3:5 → C = 8100 × 5/9 = 4500.
47. A,B started a business in which A invested Rs.10000/- for 1year, B invested Rs.20000/- for 2 years. At the end of the profit received by them is Rs.5000/-. What is A's share? a) Rs.4000/- b) Rs.3000/- c) Rs.2000/- d) Rs.1000/-	**d) Rs.1000/-** Capital × time: 10000 : 40000 = 1:4 → A = 5000/5 = 1000.
48. A,B and C started a business in which A invested Rs.10000/- for 1year, B invested Rs.20000/- for 2 years, C invested Rs.30000/- for 3 years. At the end of the profit received by them is Rs.5600/-. What is C's share? a) Rs.1600/- b) Rs.400/- c) Rs.3600/- d) Rs.2000/-	**c) Rs.3600/-** Ratio 1:4:9 → C = 5600 × 9/14 = 3600.
49. P , Q, R enter into a partnership & their share are in the ratio 1/2 : 1/3 : 1/4 , after two months , P withdraws half of the capitals & after 10 months , a profit of Rs 378 is divided among them . What is Q's share? a) 144 b) 154 c) 164 d) 174	**a) 144** Capitals 6:4:3; P = 6×2 + 3×10 = 42, Q = 48, R = 36 → Q = 378 × 48/126 = 144.
50. If 4 (P's Capital) = 6 (Q's Capital) = 10 (R's Capital) , then out of the total profit of Rs 4650 , R will receive a) 700 b) 900 c) 600 d) 750	**b) 900** P:Q:R = 1/4 : 1/6 : 1/10 = 15:10:6 → R = 4650 × 6/31 = 900.
51. In a business, A and C invested amounts in the ratio 2 : 1 , whereas the ratio between amounts invested by A and B was 3 : 2 . If Rs 157300 was their profit, how much amount did B receive? a) 48600 b) 48400 c) 48500 d) 44800	**b) 48400** A:B:C = 6:4:3 → B = 157300 × 4/13 = 48400.
53. A invested Rs 76000 in a business. After few months, B joined him with Rs 57000. The total profit was divided between them in the ratio 2 : 1 at the end of the year. After how many months did B join? a) 3 b) 4 c) 5 d) 6	**b) 4** 76000×12 : 57000×(12 – x) = 2:1 → 12 – x = 8 → x = 4.

Time & Work

Question	Answer
81. A and B together can complete a task in 7 days. B alone can do it in 20 days. What part of the work was carried out by A? A completed the job alone after A and B worked together for 5 days. Part of the work done by A could have been done by B and C together in 6 days. a) I alone sufficient while II alone not sufficient to answer b) II alone sufficient while I alone not sufficient to answer c) Either I or II alone sufficient to answer d) Both I and II are not sufficient to answer e) Both I and II are necessary to answer	**a) I alone sufficient while II alone not sufficient to answer** With I: A's rate = 1/7 − 1/20; A's part = 5 × 13/140 + 2/7 = 3/4. II involves unknown C.

Question	Answer
82. A alone can do a piece of work in 6 days and B alone in 8 days. A and B undertook to do it for Rs. 3200. With the help of C, they completed the work in 3 days. How much is to be paid to C? a) Rs. 1000 b) Rs. 400 c) Rs. 600 d) Rs. 800	**b) Rs. 400** In 3 days A does 1/2, B 3/8 → C does 1/8; C gets 3200/8 = 400.
83. A takes twice as much time as B or thrice as much time as C to finish a piece of work. Working together, they can finish the work in 2 days. B can do the work alone in: a) 4 days b) 6 days c) 8 days d) 12 days	**b) 6 days** If A takes x: 1/x + 2/x + 3/x = 1/2 → x = 12; B = 6 days.
84. A and B can complete a work in 15 days and 10 days respectively. They started doing the work together but after 2 days B had to leave and A alone completed the remaining work. The whole work was completed in : a) 8 days b) 10 days c) 12 days d) 15 days	**c) 12 days** 2 days together = 1/3; A does 2/3 in 10 days; total 12.
103. In how many days can 10 women finish a work? I. 10 men can complete the work in 6 days. II. 10 men and 10 women together can complete the work in 3 3 days 7 III. If 10 men work for 3 days and thereafter 10 women replace them, the remaining work in completed in 4 days. a) Any two of the three b) I and II only c) II and III only d) I and III only e) None of the mentioned	**a) Any two of the three** Each pair gives 10 women = 1/8 of the work per day (e.g. I+II: 7/24 − 1/6 = 1/8) → 8 days.
111. 4. How many workers are required for completing the construction work in 10 days? I. 20% of the work can be completed by 8 workers in 8 days. II. 20 workers can complete the work in 16 days. III. One-eighth of the work can be completed by 8 workers in 5 days. a) I only b) II and III only c) III only d) I and III only e) Any one of the three	**e) Any one of the three** Each statement alone gives 320 worker-days for the job → 32 workers.
112. 8 men and 14 women are working together in a field. After working for 3 days, 5 men and 8 women leave the work. How many more days will be required to complete the work? I. 19 men and 12 women together can complete the work in 18 days. II. 16 men can complete two-third of the work in 16 days. III. In 1 day, the work done by three men in equal to the work done by four women. a) I only b) II only c) III only d) I or II or III e) II or III only	**d) I or II or III** Answer as per the given key.
113. A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is : a) 1/4 b) 1/10 c) 7/15 d) 8/15	**d) 8/15** 4 × (1/15 + 1/20) = 7/15 done → 8/15 left.
114. A can lay railway track between two given stations in 16 days and B can do the same job in 12 days. With help of C, they did the job in 4 days only. Then, C alone can do the job in: a) 46/5 days b) 47/5 days c) 48/5 days d) 10 days	**c) 48/5 days** C = 1/4 − 1/16 − 1/12 = 5/48 → 48/5 days.
115. B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day? a) 12 days b) 15 days c) 16 days d) 18 days	**b) 15 days** Per 3 days: 2/20 + 1/10 = 1/5 → 5 cycles = 15 days.

Question	Answer
116. A is thrice as good as workman as B and therefore is able to finish a job in 60 days less than B. Working together, they can do it in: a) 20 days b) 45/2 days c) 25 days d) 30 days	**b) 45/2 days** 3x − x = 60 → A = 30, B = 90; together 1/30 + 1/90 = 2/45 → 45/2 days.
119. A group of workers can do a piece of work in 24 days. However as 7 of them were absent it took 30 days to complete the work. How many people actually worked on the job to complete it? a) 20 b) 24 c) 28 d) 30	**c) 28** 24n = 30(n − 7) → n = 35; worked = 28.
120. X alone can do a piece of work in 15 days and Y alone can do it in 10 days. X and Y undertook to do it for Rs. 720. With the help of Z they finished it in 5 days. How much is paid to Z? a) 120 b) 150 c) 180 d) 240	**a) 120** In 5 days X does 1/3, Y 1/2 → Z does 1/6 → 720/6 = 120.
121. A can do a certain job in 12 days. B is 60% more efficient than A . The number of days, it takes B to do the same piece of work is a) 6.5 days b) 7.5 days c) 8.5 days d) 9.5 days	**b) 7.5 days** 12 ÷ 1.6 = 7.5 days.
155. If Worker W1 can finish a job in 3 hours and W2 can finish a job in 12 hours, in how many minutes could they finish the job if they worked on it together, at their respective rates? a) 120 min b) 144 min c) 140 min d) 136 min	**b) 144 min** 1/3 + 1/12 = 5/12 per hour → 12/5 h = 144 min.
156. Factory f1 makes toys at the rate of 60 every 15 minutes. Factory f2 makes toys at the rate of 50 every 10 minutes. If the two factories begin making toys at the same moment, how many minutes will it take them to make 189 toys? a) 18 min b) 21 min c) 25 min d) 28 min	**b) 21 min** 4 + 5 = 9 toys/min → 189/9 = 21 min.

Pipes & Cistern

Question	Answer
117. Three pipes A, B and C can fill a tank from empty to full in 30 minutes, 20 minutes, and 10 minutes respectively. When the tank is empty, all the three pipes are opened. A, B and C discharge chemical solutions P,Q and R respectively. What is the proportion of the solution R in the liquid in the tank after 3 minutes? a) 5/11 b) 6/11 c) 7/11 d) 8/11	**b) 6/11** In 3 min: P = 1/10, Q = 3/20, R = 3/10 → R share = 0.30/0.55 = 6/11.
118. A tank is fitted with 12 pipes, some of them that fill the tank and others that are waste pipe meant to empty the tank. Each of the pipes that fill the tank can fill it in 8 hours, while each of those that empty the tank can empty it in 24 hours. If all the pipes are kept open when the tank is full, it will take exactly 6 hours for the tank to empty. How many of these are fill pipes? a) 6 b) 7 c) 8 d) 9	**b) 7** Answer as per the given key.

Question	Answer
122. Two taps A and B can fill a tank in 10 hours and 15 hours respectively. If both the taps are opened together, in how many hours will the tank will be full ? a) 6 hours b) 7 hours c) 8 hours d) 9 hours	**a) 6 hours** 1/10 + 1/15 = 1/6 → 6 hours.
123. Two pipes can fill a tank in 10 hours and 12 hours resp. while third pipe empties the full tank in 20 hours. If all the three pipes operate simultaneously, in how much time the tank will be filled? (in hours) a) 7.5 b) 6.5 c) 10.5 d) 9.5	**a) 7.5** 1/10 + 1/12 − 1/20 = 8/60 → 7.5 hours.
124. A pipe can fill a cistern in 7 hours. Due to a leak in the bottom, the cistern is full in 8 hours. The leak can empty the cistern is: a) 56 hours b) 1 hour c) 20 hours d) 15 hours	**a) 56 hours** Leak = 1/7 − 1/8 = 1/56 → 56 hours.
125. An electric pump can fill a tank in 3 hours. Because of a leak in the tank, it took 3 ½ hours to fill the tank. The leak can drain out of all the water of the tank in a) 17 hours b) 15 hours c) 21 hours d) 23 hours	**c) 21 hours** Leak = 1/3 − 2/7 = 1/21 → 21 hours.
128. Pipes A and B can fill a tank in 5 and 6 hours respectively. Pipe C can empty it in 12 hours. If all the three pipes are opened together, then the tank will be filled in: a) 60/17 hours b) 30/11 hours c) 30/17 hours d) 90/17 hours	**a) 60/17 hours** 1/5 + 1/6 − 1/12 = 17/60 → 60/17 hours.
140. A tank is filled by three pipes with uniform flow. The first two pipes operating simultaneously fill the tank in the same time during which the tank is filled by the third pipe alone. The second pipe fills the tank 5 hours faster than the first pipe and 4 hours slower than the third pipe. The time required by the first pipe is: a) 6 hours b) 10 hours c) 15 hours d) 20 hours	**c) 15 hours** 1/x + 1/(x−5) = 1/(x−9) → $x^2 - 18x + 45 = 0$ → x = 15.
150. Two pipes can fill a tank in 20 and 24 minutes respectively and a waste pipe can empty 3 gallons per minute. All the three pipes working together can fill the tank in 15 minutes. The capacity of the tank is: a) 60 gallons b) 90 gallons c) 100 gallons d) 120 gallons	**d) 120 gallons** Waste = 1/20 + 1/24 − 1/15 = 1/40 → empties in 40 min × 3 gal = 120 gallons.
152. A tank is filled in 5 hours by three pipes A, B and C. The pipe C is twice as fast as B and B is twice as fast as A. How much time will pipe A alone take to fill the tank? a) 20 hours b) 30 hours c) 35 hours d) 40 hours	**c) 35 hours** Rates a, 2a, 4a: 7a = 1/5 → A alone = 35 hours.
153. Two pipes A and B together can fill a cistern in 4 hours. Had they been opened separately, then B would have taken 6 hours more than A to fill the cistern. How much time will be taken by A to fill the cistern separately? a) 1 hour b) 2 hours c) 6 hours d) 8 hours	**c) 6 hours** 1/x + 1/(x+6) = 1/4 → $x^2 - 2x - 24 = 0$ → x = 6.
154. Two pipes A and B can fill a tank in 20 and 30 minutes respectively. If both the pipes are used together, then how long will it take to fill the tank? a) 12 min b) 15 min c) 10 min d) 18 min	**a) 12 min** 1/20 + 1/30 = 1/12 → 12 min.

Speed

Question	Answer
157. If Kathy can drive a distance of 50 miles in 70 minutes, how many minutes in terms of x, will it take for her to drive 120 miles at the same speed? a) 150 min b) 164 min c) 168 min d) 172 min	**c) 168 min** 70/50 × 120 = 168 min.
168. A racing car spots a cyclist at a distance of 100 m. It overtakes the cyclist and then cyclist can observe the racing car at a distance of 200 m. If the car's speed is 5 times that of the cyclist and the time lapse between the moment the car racer spots the cyclist and the last moment the cyclist observes the racer is 5 sec, the speed of the car racer is: a) 15 m/s b) 60 m/s c) 75 m/s d) 100 m/s	**c) 75 m/s** Relative speed 4v covers 300 m in 5 s → v = 15; car = 75 m/s.
192. Watson and Sherlock drove to a place 120 miles away in the same car. Watson drove to the place and Sherlock drove back along the same route. If Watson drove at 60 miles per hour and Sherlock drove at 50 miles per hour, how much longer, in minutes, did it take Sherlock to travel the distance than it did Watson? a) 21 min b) 22 min c) 23 min d) 24 min	**d) 24 min** 120/50 − 120/60 = 0.4 h = 24 min.
194. If a man walks at the rate of 5kmph, he misses a train by only 7min. However if he walks at the rate of 6 kmph he reaches the station 5 minutes before the arrival of the train. Find the distance covered by him to reach the station. a) 6 km b) 7 km c) 8 km d) 9 km	**a) 6 km** d/5 − d/6 = 12/60 → d = 6 km.

Problems on Trains

Question	Answer
158. A train 100 m long is running at the speed of 30 km / hr. Find the time taken byit to pass a man standing near the railway line. a) 10 sec b) 11 sec c) 12 sec d) 13 sec	**c) 12 sec** 30 km/h = 25/3 m/s → 100 ÷ 25/3 = 12 s.
159. A train is moving at a speed of 132 km/br. If the length of the train is 110 metres, how long will it take to cross a railway platform 165 metres long? a) 14/2 sec b) 15/2 sec c) 16/2 sec d) 17/2 sec	**b) 15/2 sec** 132 km/h = 110/3 m/s; 275 ÷ 110/3 = 7.5 s.
160. A man is standing on a railway bridge which is 180 m long. He finds that a train crosses the bridge in 20 seconds but himself in 8 seconds. Find the length of the train and its speed? a) 120 km and 54 kmph b) 120 km and 55 kmph c) 110 km and 75 kmph d) 110 km and 85 kmph	**a) 120 km and 54 kmph** Speed = 180/(20 − 8) = 15 m/s = 54 kmph; length = 15 × 8 = 120 m.
161. A train 150 m long is running with a speed of 68 kmph. In what time will it pass a man who is running at 8 kmph in the same direction in which the train is going? a) 7 sec b) 8 sec c) 9 sec d) 10 sec	**c) 9 sec** Relative 60 km/h = 50/3 m/s → 150 ÷ 50/3 = 9 s.

Question	Answer
162. A train 220 m long is running with a speed of 59 kmph. In what time will it pass a man who is running at 7 kmph in the direction opposite to that in which the train is going? a) 11 sec b) 12 sec c) 13 sec d) 14 sec	**b) 12 sec** Relative 66 km/h = 55/3 m/s → 220 ÷ 55/3 = 12 s.
163. Two trains 137 metres and 163 metres in length are running towards each other on parallel lines, one at the rate of 42 kmph and another at 48 kmpb. In what time will they be clear of each other from the moment they meet? a) 11 sec b) 12 sec c) 13 sec d) 14 sec	**b) 12 sec** Relative 90 km/h = 25 m/s → 300 ÷ 25 = 12 s.
164. Two trains 100 metres and 120 metres long are running in the same direction with speeds of 72 km/hr,In howmuch time will the first train cross the second? a) 33 sec b) 44 sec c) 55 sec d) 66 sec	**b) 44 sec** Answer as per the given key (the second train's speed is missing; 44 s fits 72 and 54 km/h).
165. A train 100 metres long takes 6 seconds to cross a man walking at 5 kmph in the direction opposite to that of the train. Find the speed of the train.? a) 33 kmph b) 44 kmph c) 55 kmph d) 66 kmph	**c) 55 kmph** Relative = 100/6 m/s = 60 km/h → train = 60 − 5 = 55 kmph.
166. A train running at 54 kmph takes 20 seconds to pass a platform. Next it takes.12 sec to pass a man walking at 6 kmph in the same direction in which the train is going . Find the length of the train and the length of the platform. a) 100 m and 200 m b) 140 m and 140 m c) 160m and 140 m d) 140 m and 160 m	**c) 160m and 140 m** Relative 48 km/h × 12 s = 160 m train; 15 m/s × 20 s = 300 → platform 140 m.
167. A man sitting in a train which is traveling at 50 kmph observes that a goods train, traveling in opposite direction, takes 9 seconds to pass him. If the goods train is 280 m long, find its speed.? a) 59 kmph b) 60 kmph c) 61 kmph d) 62 kmph	**d) 62 kmph** Relative = 280/9 m/s = 112 km/h → 112 − 50 = 62 kmph.
179. A train is moving at speed of 132kmph. If the length of the train is 110 meters, how long will it take to cross a railway platform 165 m long?(in sec) a) 7 b) 5 c) 7.5 d) None of the mentioned.	**c) 7.5** 275 m ÷ 110/3 m/s = 7.5 s.
190. Two trains 200mts and 150mts are running on the parallel rails at the rates of 40km/hr and 45km/hr. In how much time will they cross each other if they are running in the same direction. a) 248 sec b) 250 sec c) 252 sec d) None of the mentioned	**c) 252 sec** Relative 5 km/h = 25/18 m/s → 350 ÷ 25/18 = 252 s.
193. Two trains are at stations 90 miles apart and they start running towards each other at the exact same moment. If the speed for Train 1 is 50 miles per hour and Train 2 is 40 miles per hour, how many miles has Train 1 travelled when it meets Train 2? a) 30 b) 40 c) 50 d) 60	**c) 50** They meet after 90/90 = 1 h → Train 1 = 50 miles.

Question	Answer
195. Train A traveling at 60 km/hr leaves Mumbai for Delhi at 6 P.M. Train B traveling at 90 km/hr also leaves Mumbai for Delhi at 9 P.M. Train C leaves Delhi for Mumbai at 9 P.M. If all three trains meet at the same time between Mumbai and Delhi, what is the speed of Train C if the distance between Delhi and Mumbai is 1260 kms? a) 100 kmph b) 120 kmph c) 150 kmph d) 180 kmph	**b) 120 kmph** B catches A 180/30 = 6 h after 9 PM, 540 km out; C covers 720 km in 6 h = 120 kmph.
197. A train is moving at a speed of 132 km/hour. If the length of the train is 110 meters, how long will it taketo cross a railway platform 165 meters long. a) 7s b) 7.5s c) 8s d) 8.5s	**b) 7.5s** 275 m ÷ 110/3 m/s = 7.5 s.
198. Two trains running in opposite directions cross a man standing on the platform in 27 seconds and 17seconds respectively and they cross each other in 23 seconds. The ratio of their speeds is ? a) 1:3 b) 3:2 c) 3:5 d) 5:7	**b) 3:2** 27x + 17y = 23(x + y) → 4x = 6y → x:y = 3:2.
208. In what time will a train 100 meters long cross an electric pole, if its speed is 144 km/hr a) 5s b) 4.5s c) 3s d) 2.5s	**d) 2.5s** 144 km/h = 40 m/s → 100/40 = 2.5 s.
219. How long does a train 110 meters long running at the speed of 72 km/hour take to cross a bridge 132meters in length ? a) 15s b) 12.1s c) 10 d) 8.1s	**b) 12.1s** 242 m ÷ 20 m/s = 12.1 s.
221. A train is 360 meter long is running at a speed of 45 km/hour. In what time will it pass a bridge of 140meter length. a) 20s b) 30s c) 40s d) 50s	**c) 40s** 500 m ÷ 12.5 m/s = 40 s.
222. A train running at the speed of 60 km/hr crosses a pole in 9 seconds. Find the length of the train. a) 150 meter b) 145 meter c) 140 meter d) 135 meter	**a) 150 meter** 60 km/h = 50/3 m/s × 9 = 150 m.
223. Length of train is 130 meters and speed of train is 45 km/hour. This train can pass a bridge in 30seconds, then find the length of the bridge. a) 230 meters b) 235 meters c) 240 meters d) 245 meters	**d) 245 meters** 12.5 m/s × 30 = 375 m → bridge = 375 − 130 = 245 m.
224. Speed of a goods train is 72 km/hr. This train crosses a 250 meter platform in 26 seconds. Then findthe length of goods train. a) 250 meters b) 260 meters c) 270 meters d) 280 meters	**c) 270 meters** 20 m/s × 26 = 520 m → train = 520 − 250 = 270 m.
225. A 300 meter long train crosses a platform in 39 seconds while it crosses a signal pole in 18 seconds.What is the length of the platform. a) 310 meters b) 335 meters c) 345 meters d) 350 meters	**d) 350 meters** Speed = 300/18 = 50/3 m/s; 39 s → 650 m → platform = 350 m.
226. A train speeds past a pole in 15 seconds and a platform 100 meter long in 25 seconds. What is lengthof the train ? a) 140 meters b) 145 meters c) 150 meters d) 155 meters	**c) 150 meters** L/15 = (L + 100)/25 → L = 150 m.

Problems on Boats & Streams

Question	Answer
227. A man can row upstream 10 kmph and downstream 20 kmph. Find the man rate in still water and rate of the stream. a) 0, 5 b) 5, 5 c) 15, 5 d) 10, 5	**c) 15, 5** Still = (20 + 10)/2 = 15; stream = (20 − 10)/2 = 5.
228. If a boat goes 7 km upstream in 42 minutes and the speed of the stream is 3 kmph, then the speed of the boat in still water is a) 12 kmph b) 13 kmph c) 14 kmph d) 15 kmph	**b) 13 kmph** Upstream = 7 ÷ 0.7 h = 10 → boat = 10 + 3 = 13.
229. A man can row at 5 kmph in still water. If the velocity of the current is 1 kmph and it takes him 1 hour to row to a place and come back. how far is that place. a) 0.4 km b) 1.4 km c) 2.4 km d) 3.4 km	**c) 2.4 km** d/6 + d/4 = 1 → d = 2.4 km.
230. The speed of a boat in still water is 15 km/hr and the rate of current is 3 km/hr. The distance travelled downstream in 12 minutes is a) 1.6 km b) 2 km c) 3.6 km d) 4 km	**c) 3.6 km** 18 km/h × 12/60 h = 3.6 km.
231. Sahil can row 3 km against the stream in 20 minutes and he can return in 18 minutes. What is rate of current ? a) 1/2 kmph b) 1/3 kmph c) 2 kmph d) 4 kmph	**a) 1/2 kmph** Up = 9, down = 10 → current = (10 − 9)/2 = 1/2 kmph.
238. A man takes 3 hours 45 minutes to row a boat 15 km downstream of a river and 2 hours 30 minutes to cover a distance of 5 km upstream. Find the speed of the current. a) 1 kmph b) 2 kmph c) 3 kmph d) 4 kmph	**a) 1 kmph** Down = 15/3.75 = 4, up = 5/2.5 = 2 → current = 1.
249. In one hour, a boat goes 11km along the stream and 5 km against it. Find the speed of the boat in still water. a) 6 kmph b) 7 kmph c) 8 kmph d) 9 kmph	**c) 8 kmph** (11 + 5)/2 = 8.
251. A motorboat, whose speed in 15 km/hr in still water goes 30 km downstream and comes back in a total of 4 hours 30 minutes. The speed of the stream (in km/hr) is a) 2 kmph b) 3 kmph c) 4 kmph d) 5 kmph	**d) 5 kmph** 30/(15+s) + 30/(15−s) = 4.5 → 225 − s^2 = 200 → s = 5.
252. If Rahul rows 15 km upstream in 3 hours and 21 km downstream in 3 hours, then the speed of the stream is a) 5 kmph b) 4 kmph c) 2 kmph d) 1 kmph	**d) 1 kmph** Up 5, down 7 → stream = 1.
253. A man rows 750 m in 675 seconds against the stream and returns in 7 and half minutes. His rowing speed in still water is a) 4 kmph b) 5 kmph c) 6 kmph d) 7 kmph	**b) 5 kmph** Up = 4 km/h, down = 6 km/h → still = 5.
254. A boat can travel with a speed of 16 km/hr in still water. If the rate of stream is 5 km/hr, then find the time taken by the boat to cover distance of 84 km downstream. a) 4 hours b) 5 hours c) 6 hours d) 7 hours	**a) 4 hours** 84 ÷ 21 = 4 hours.
255. A man's speed with the current is 20 kmph and speed of the current is 3 kmph. The Man's speed against the current will be a) 11 kmph b) 12 kmph c) 14 kmph d) 17 kmph	**c) 14 kmph** Still = 17 → against = 17 − 3 = 14.

Question	Answer
256. A man takes twice as long to row a distance against the stream as to row the same distance in favour of the stream. The ratio of the speed of the boat in still water and stream is a) 3:1 b) 1:3 c) 2:4 d) 4:2	**a) 3:1** b + s = 2(b − s) → b = 3s → 3:1.

Allegation & Mixtures

Question	Answer
257. A can contains a mixture of two liquids A and B in the ratio 7 : 5. When 9 litres of mixture are drawn off and the can is filled with B, the ratio of A and B becomes 7 : 9. How many litres of liquid A was contained by the can initially? a) 10 b) 20 c) 21 d) 25	**c) 21** (7x − 21/4)/(5x + 21/4) = 7/9 → x = 3 → A = 21 L.
258. A milk man sells the milk at the cost price but he mixes the water in it and thus he gains 9.09%. The quantity of water in the mixture of 1 liter is : a) 83.33 ml b) 89.93 ml c) 90.87 ml d) 82.67 ml	**a) 83.33 ml** Gain 1/11 → water : milk = 1 : 11 → 1000/12 = 83.33 ml.
268. 8 litres are drawn from a cask full of wine and is then filled with water. This operation is performed three more times. The ratio of the quantity of wine now left in cask to that of the water is 16 : 65. How much wine the cask hold originally? a) 18 litres b) 20 litres c) 24 litres d) 15 litres	**c) 24 litres** $(1 - 8/x)^4 = 16/81 \rightarrow 1 - 8/x = 2/3 \rightarrow x = 24$.
271. A vessel is filled with liquid, 3 parts of which are water and 5 parts of syrup. How much of the mixture must be drawn off and replaced with water so that the mixture may be half water and half syrup? a) 1/3 b) 1/5 c) 1/2 d) 1/8	**b) 1/5** Syrup (5/8)(1 − f) = 1/2 → f = 1/5.
273. A container contains 40 litres of milk.From this container 4 litres of milk was taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container. a) 29.16 litres b) 24.23 litres c) 26.78 litres d) 21.98 litres	**a) 29.16 litres** $40 \times (36/40)^3 = 29.16$ L.
274. A mixture of 150 liters of wine and water contains 20% water. How much more water should be added so that water becomes 25% of the new mixture? a) 10 litres b) 15 litres c) 20 litres d) 12 litres	**a) 10 litres** (30 + x)/(150 + x) = 1/4 → x = 10.
275. The ratio of petrol and kerosene in the container is 3:2 when 10 liters of the mixture is taken out and is replaced by the kerosene, the ratio become 2:3. Then total quantity of the mixture in the container is: a) 30 b) 25 c) 40 d) 50	**a) 30** 3(3T/5 − 6) = 2(2T/5 + 6) → T = 30 L.
276. How many kilograms of sugar costing Rs. 9 per kg must be mixed with 27 kg of sugar costing Rs. 7 per Kg so that there may be a gain of 10 % by selling the mixture at Rs. 9.24 per Kg ? a) 63 kg b) 34 kg c) 78 kg d) 38 kg	**a) 63 kg** Mixture CP = 9.24/1.1 = 8.40; alligation 1.40 : 0.60 = 7:3 → 27 × 7/3 = 63 kg.